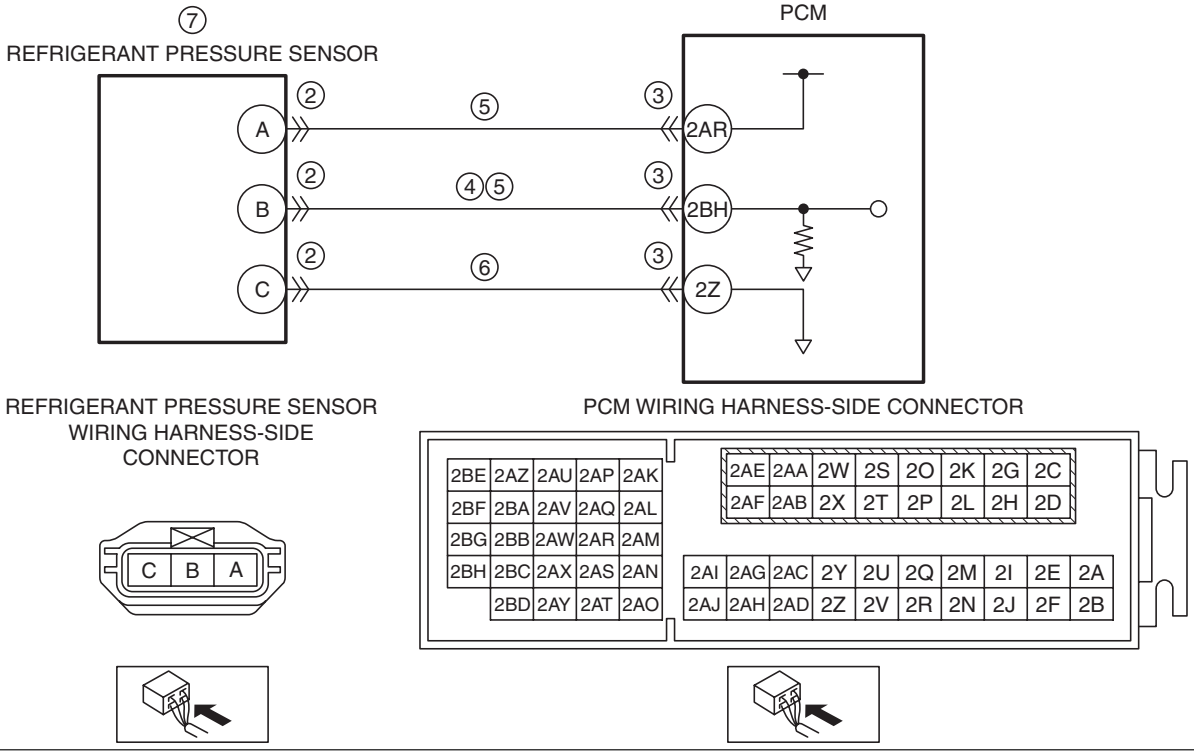
DTC P0533:00 [SKYACTIV-D 1.5]

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DTC P0533:00	Refrigerant pressure sensor circuit high input
DETECTION CONDITION	• The PCM monitors the input voltage from the refrigerant pressure sensor. If the input voltage at the PCM terminal 2BH is **above 4.95 V** for **5 s**, the PCM determines that the refrigerant pressure sensor circuit has a malfunction. **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	• Inhibits the A/C control.
POSSIBLE CAUSE	• Refrigerant pressure sensor connector or terminals malfunction • PCM connector or terminals malfunction • Short to power supply in wiring harness between refrigerant pressure sensor terminal B and PCM terminal 2BH • Refrigerant pressure sensor power supply circuit and signal circuit are shorted to each other • Open circuit in wiring harness between refrigerant pressure sensor terminal C and PCM terminal 2Z • Refrigerant pressure sensor malfunction • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**INSPECT REFRIGERANT PRESSURE SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the refrigerant pressure sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.
4	**INSPECT REFRIGERANT PRESSURE SENSOR SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the refrigerant pressure sensor and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the refrigerant pressure sensor terminal B (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between refrigerant pressure sensor terminal B and PCM terminal 2BH. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 8.
5	**INSPECT REFRIGERANT PRESSURE SENSOR POWER SUPPLY CIRCUIT AND SIGNAL CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the refrigerant pressure sensor and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between refrigerant pressure sensor terminals A and B (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Refrigerant pressure sensor terminal A—PCM terminal 2AR • Refrigerant pressure sensor terminal B—PCM terminal 2BH **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 8.
		No	Go to the next step.
6	**INSPECT REFRIGERANT PRESSURE SENSOR GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the refrigerant pressure sensor and PCM connectors are disconnected. • Inspect for continuity between refrigerant pressure sensor terminal C (wiring harness-side) and PCM terminal 2Z (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between refrigerant pressure sensor terminal C and PCM terminal 2Z. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 8.
7	**INSPECT REFRIGERANT PRESSURE SENSOR** • Reconnect all disconnected connectors. • Inspect the refrigerant pressure sensor. (See REFRIGERANT PRESSURE SENSOR INSPECTION [MANUAL AIR CONDITIONER].) (See REFRIGERANT PRESSURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is there any malfunction?	Yes	Replace the refrigerant pressure sensor, then go to the next step. (See REFRIGERANT PRESSURE SENSOR REMOVAL/ INSTALLATION [MANUAL AIR CONDITIONER].) (See REFRIGERANT PRESSURE SENSOR REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Start the engine. • Turn the A/C switch on. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
9	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.